

Tiago O. e Nicollas C.

```
#include <Servo.h>
```

```
int valor;
```

```
long Centimetro;
```

```
Servo servo1;
```

```
long readUltrasonicDistance(int triggerPin, int echoPin)
```

```
{
```

```
pinMode(triggerPin, OUTPUT);
```

```
digitalWrite(triggerPin, LOW);
```

```
delayMicroseconds(2);
```

```
digitalWrite(triggerPin, HIGH);
```

```
delayMicroseconds(10);
```

```
digitalWrite(triggerPin, LOW);
```

```
pinMode(echoPin, INPUT);
```

```
return pulseIn(echoPin, HIGH);
```

```
}
```

```
void setup()
```

```
{
```

```
randomSeed(analogRead(0));
```

```
pinMode(11, OUTPUT);
```

```
pinMode(10, OUTPUT);
```

```
pinMode(9, OUTPUT);  
pinMode(8, OUTPUT);  
pinMode(7, OUTPUT);  
pinMode(6, OUTPUT);  
pinMode(5, OUTPUT);
```

```
pinMode(2, INPUT);
```

```
servo1.attach(12);
```

```
servo1.write(0);  
}
```

```
void loop()  
{
```

```
Centimetro = 0.01723 * readUltrasonicDistance(A0, A0);
```

```
if (Centimetro < 40)  
{  
    sorteia();
```

```
    delay(2000);
```

```
    if (valor % 2 == 0)  
    {  
        servo1.write(90);  
        delay(1000);
```

```
        servo1.write(0);
```

```
}
```

```
digitalWrite(11, LOW);
```

```
digitalWrite(10, LOW);
```

```
digitalWrite(9, LOW);
```

```
digitalWrite(8, LOW);
```

```
digitalWrite(7, LOW);
```

```
digitalWrite(6, LOW);
```

```
digitalWrite(5, LOW);
```

```
while (Centimetro < 40)
```

```
{
```

```
Centimetro = 0.01723 * readUltrasonicDistance(A0, A0);
```

```
delay(100);
```

```
}
```

```
}
```

```
}
```

```
void sorteia()
```

```
{
```

```
valor = random(1, 7);
```

```
digitalWrite(11, LOW);
```

```
digitalWrite(10, LOW);
```

```
digitalWrite(9, LOW);
```

```
digitalWrite(8, LOW);
```

```
digitalWrite(7, LOW);
```

```
digitalWrite(6, LOW);
```

```
digitalWrite(5, LOW);
```

```
    if (valor == 1)
    {
digitalWrite(8, HIGH);
    }
```

```
    else if (valor == 2)
    {
digitalWrite(5, HIGH);
digitalWrite(9, HIGH);
    }
```

```
    else if (valor == 3)
    {
digitalWrite(8, HIGH);
digitalWrite(5, HIGH);
digitalWrite(9, HIGH);
    }
```

```
    else if (valor == 4)
    {
digitalWrite(5, HIGH);
digitalWrite(9, HIGH);
digitalWrite(11, HIGH);
digitalWrite(7, HIGH);
    }
```

```
    else if (valor == 5)
    {
```

```
digitalWrite(11, HIGH);  
digitalWrite(9, HIGH);  
digitalWrite(5, HIGH);  
digitalWrite(8, HIGH);  
digitalWrite(7, HIGH);  
    }
```

```
    else if (valor == 6)  
    {  
digitalWrite(11, HIGH);  
digitalWrite(10, HIGH);  
digitalWrite(9, HIGH);
```

```
  
digitalWrite(5, HIGH);  
digitalWrite(7, HIGH);  
digitalWrite(6, HIGH);  
    }  
}
```

